

When Do Prediction Markets Beat Static Pools? A Controlled Stress Test of LLM Belief Aggregation

Hrishi Kabra

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Abstract

We build a synthetic information environment whose Bayes-optimal posterior is known in closed form, and use it to measure how much of the available information two aggregation mechanisms recover from a pool of LLM agents: independent static pooling and dynamic LMSR market trading. Each question is elicited twice from the same agents on the same evidence — once in isolation (feeding all static baselines) and once through three rounds of market trading — so mechanism, model knowledge, and signal structure are separated by design. On the main grid the market **underperforms** static pools of the same agents’ independent beliefs: the final price has a larger posterior gap than every pool we tried, reversing the direction one might naively expect and consistent with the pre-registration’s allowance for a reversed result. A mixed-model team beats its homogeneous counterparts after quality matching (a replication of the diversity direction from prior monoculture work). An adversary instructed to push the wrong side moves prices and degrades accuracy, but loses money at every bankroll multiple — the market taxes manipulation even as some final predictions flip. Persistent wealth across questions slightly worsens accuracy. An exploratory extension along a model-capability axis — added after the initial release and not pre-registered — finds this deficit shrinks to parity as trader capability rises, with no tested tier’s market beating its own independent pool, and a same-family control shows the change is trading skill rather than added knowledge. The main experiment cost \$0.73; the capability extension added \$3.06.

1. Introduction and positioning

Prediction markets are an old idea for aggregating dispersed private information, and a recent line of work has shown that LLM agents can be made to trade in them. Two facts are by now established in the literature and are *not* what this paper claims. First, LLM agents with dispersed private signals can trade via a logarithmic market scoring rule (LMSR) and produce a price that reflects pooled information (arXiv:2604.20050, the closest prior work). Second, pools of same-family LLMs have highly correlated errors, and preference optimization can drive that correlation up to the point that ten agents behave like roughly one (arXiv:2606.26583; see also arXiv:2506.07962 and arXiv:2605.00844 on cross-model error correlation).

What that prior work entangles is the *decomposition*. When a market of LLM agents lands near or far from the truth, at least four things are moving at once: (i) how much the models know, (ii) how the private signals are structured and how informative their union is, (iii) how the agents behave under a trading prompt versus a plain elicitation prompt, and (iv) the market dynamics themselves. If you only compare mechanisms against each other, or against realized outcomes, you cannot say whether a market helped because trading aggregated information or merely because the underlying models were good, nor whether a static pool lost because pooling is weak or because the signals

were nearly redundant.

This paper is a **controlled decomposition and stress test**, not a first demonstration. The contribution is separation, built from four design choices:

1. A **generative environment with an exact Bayes-posterior oracle**. Because we generate the latent state and the signals ourselves, we can compute the Bayes-optimal posterior given any subset of signals in closed form. Every mechanism is scored against that exact target (posterior gap), not only against the realized binary outcome (Brier).
2. An **independent-elicitation control**. Each agent states a belief in isolation before any market opens, on exactly the evidence it will later trade on. All static baselines are computed from these Pass-1 beliefs, so the market and the pools consume identical information from identical agents.
3. **Quality-matched team comparison**. The homogeneous and mixed teams are matched on standalone posterior gap (measured on held-out pilot questions), so a mixed team’s advantage cannot be attributed to simply swapping in a stronger model.
4. The resulting **map of where the market helps, does nothing, or hurts**, read against the Bayes optimum rather than against the market’s own past.

The deliverable is that map, plus a stress test of the market under adversarial capital and under wealth persistence across questions. We pre-registered the hypotheses and elicitation protocol (SPEC.md) before running any main experiment; directions were left open where we genuinely did not know them.

2. The generative environment

The environment is the scientific asset, so we describe the formal model first and the prose rendering second.

Latent state and signals. Each question has a binary latent state $\mathbf{s} \in \{0,1\}$ — the answer — with a uniform prior, $\mathbf{s} \sim \text{Bernoulli}(0.5)$. From $\mathbf{s}$ we generate six signals through a two-channel process. A common latent channel $\mathbf{z}$ equals $\mathbf{s}$ with probability $\mathbf{a}_z$ (and $1-\mathbf{s}$ otherwise); this channel is what lets us dial correlation. Each signal i either copies the common channel ($\mathbf{x}_i = \mathbf{z}$) with probability λ_i , or, with probability $1-\lambda_i$, draws an independent private channel that equals $\mathbf{s}$ with the signal’s own accuracy $\mathbf{a}_i$. Setting all $\lambda_i = 0$ makes the six signals conditionally independent given $\mathbf{s}$, each with accuracy $\mathbf{a}_i$; raising λ_i introduces shared-error correlation. The main grid uses six conditionally independent signals ($\lambda_i = 0$), each with accuracy 0.75, and a common-channel accuracy $\mathbf{a}_z = 0.8$.

The oracle. For any observed subset $\mathbf{A}$ of signals with values $\mathbf{x}_A$, the likelihood marginalizes the common channel: $P(\mathbf{x}_A \mid \mathbf{s}) = \sum_z P(\mathbf{z} \mid \mathbf{s}) \cdot \prod_{i \in \mathbf{A}} P(\mathbf{x}_i \mid \mathbf{s}, \mathbf{z})$, and the posterior $P(\mathbf{s}=1 \mid \mathbf{x}_A)$ follows by Bayes’ rule under the uniform prior. This is computed exactly, without sampling, and stored for every signal subset. Two properties matter and are checked against the oracle at generation time, not by eyeball: the posterior given the union of all six signals is informative, and **no single shard is sufficient** — the posterior given any one signal differs from the posterior given all six by at least a configured margin of 0.1. Questions that fail either check are rejected and resampled. This guarantees that information is genuinely dispersed across agents, so that aggregation has something to do.

Prose rendering. Each signal is rendered into a natural-language shard — a witness statement or

document snippet — under a template constraint so the text entails exactly its signal and nothing more. A separate adversarial verification pass checks three things per shard: that the shard entails its signal in the correct direction (direction entailment), that it does not leak any other agent’s signal (no cross-shard leaks), and that it carries no meta-clue about its own reliability (no accuracy hints). We rendered and verified 50 questions (300 shards). The first verification round passed 80% of questions (shard-level: direction 97.3%, leak 99.0%, meta 99.0%); two surgical re-render rounds with fresh verifiers brought all 50 questions to a 100% pass rate on all three criteria. A manual spot-check of 10 shards across five questions confirmed the automated verdicts. The dataset is 40 main questions plus 10 held-out pilot questions, generated once and frozen as versioned JSONL so every condition and seed loads identical questions — paired comparisons hold by construction.

3. Mechanisms and protocol

Market. The market is Hanson’s LMSR for a binary YES/NO outcome, with cost function $C(q) = b \cdot \log(\exp(q_{\text{yes}}/b) + \exp(q_{\text{no}}/b))$ and price $p_{\text{yes}} = 1/(1 + \exp((q_{\text{no}} - q_{\text{yes}})/b))$, implemented in a log-sum-exp-stable form (Hanson 2003; Abernethy, Chen, and Wortman Vaughan 2013; Storkey 2011). The liquidity parameter b sets how much cash moves the price; pushing YES from 0.5 to 0.8 costs about $b \cdot \ln(2.5) \approx 37$, roughly a third of a 100-unit bankroll. We chose $b = 40$ by a pilot sweep over $\{20, 40, 80\}$: at $b = 20$ prices went degenerate (pinned to an extreme) on a large fraction of pilot questions, at $b = 80$ the market was sluggish and showed no advantage, and $b = 40$ was non-degenerate (degenerate on 10% of pilot questions) while remaining responsive. Each agent starts with a bankroll of 100. The market runs three rounds; agent order is reshuffled each round from a seeded RNG. Requested trade sizes are clamped to affordability server-side, and every turn — including holds — is logged. b and the prompts were frozen after the pilot.

Two-pass elicitation. This is the contamination control, followed exactly.

- **Pass 1 (independent).** Each agent sees only the question, its own shard(s), and its persona — no prices, no other agents, no mention of a market. It outputs a **Belief** (a probability plus a short rationale). These Pass-1 beliefs feed *all* static baselines and *all* error-correlation estimates.
- **Pass 2 (market).** The same agents trade on the same shards in a fresh context, each told its own Pass-1 conclusion as its prior. In-market stated beliefs are logged for dynamics analysis only and are never used as a baseline.

A code-level phase guard enforces the separation: baseline functions accept only records tagged `phase="independent"` and assert on anything else. This means the market and the static pools always aggregate the *same information from the same agents* — the only thing that differs is the mechanism.

Baselines. From the Pass-1 beliefs we compute four static pools: the mean, the median, a log opinion pool, and a leave-one-out calibrated stack. All are scored against the oracle posterior (posterior gap) and the realized outcome (Brier).

Agents. Six fixed personas — base-rate skeptic, evidence maximalist, contrarian, domain specialist, Kelly-style sizer, and momentum reader — are held constant across conditions; each agent is assigned one shard. Three models back the pools: `meta-llama/llama-3.1-8b-instruct` (model A), `deepseek/deepseek-v4-flash` (model B), and `nvidia/nemotron-3-super-120b-a12b` (model C, run on a free tier in the homogeneous arm). Their standalone posterior gaps, measured on the pilot questions, are 0.291, 0.268, and 0.257 respectively — closely matched, which is what makes the

mixed-team comparison a comparison of composition rather than of raw model quality. Structured output uses guided decoding where available and otherwise a validate-and-one-retry loop, falling back to a logged `hold` on a second failure.

Condition grid. All conditions run over 40 questions $\times$ 3 seeds. Condition A is a homogeneous deepseek pool (it also serves as homogeneous arm B2 and as the Pass-1 source for the wealth conditions). B1 and B3 are homogeneous llama and nemotron pools. C is the mixed A/B/C team, two agents per model. D adds an adversary to pool C at bankroll multiples $k \in \{1, 3, 10\}$. W runs pool A with wealth persisting across questions, in fixed and shuffled question orders. The D and W conditions reuse their source condition’s Pass-1 beliefs and run the market pass only. Main-grid spend was \$0.7255, and the whole experiment cost \$0.73 across all ledgers including the pilots.

4. Results

Every number below is quoted from the analysis pipeline’s frozen outputs (`analysis_out/stats.md`, `table1.md`, `table2.md`), reproducible from the released event logs. Figures 1 and 2 and Tables 1 and 2 live in `analysis_out/`. The primary metric is the posterior gap against the oracle; comparisons use a paired bootstrap 95% CI and a Wilcoxon signed-rank test over the 40 questions $\times$ 3 seeds, with per-seed means reported for transparency.

4.1 H1 — market vs. static pools (reversed)

H1: On info-asymmetric questions, the final market price has a smaller posterior gap than static pools computed over independently elicited pre-market beliefs from the same agents on the same shards. Direction unknown — a null or reversed result is a reportable finding.

The result is **reversed** (Fig 1, `analysis_out/fig1.png`). On condition A the market’s final price has a posterior gap of 0.269 ± 0.012 , larger than every static pool built from the same agents’ independent beliefs. Against the mean pool (gap 0.218 ± 0.009) the market is worse by $\Delta = 0.051$ (95% CI $[-0.001, 0.100]$, Wilcoxon $p = 0.080$). Against the other three pools the market is worse by a significant margin: the median pool ($\Delta = 0.081$, CI $[0.038, 0.123]$, $p = 0.005$), the log opinion pool ($\Delta = 0.060$, CI $[0.008, 0.109]$, $p = 0.048$), and the calibrated stack ($\Delta = 0.073$, CI $[0.019, 0.122]$, $p = 0.008$). Per-seed means are tight and all positive (e.g. 0.084 / 0.082 / 0.078 for the median pool), so the direction is stable across seeds.

The Brier deltas point the same way — the market’s Brier (0.281 ± 0.014) is worse than the mean pool’s (0.202 ± 0.001) by $\Delta = 0.079$ — but none of the four Brier comparisons is individually significant (all $p \geq 0.085$). The headline is that in this regime, LMSR trading **destroys** information relative to simply pooling the same agents’ independent beliefs. This is a reversal of the naive expectation and squarely within what the pre-registration flagged as a reportable outcome.

Two caveats bound this result. It holds for **this regime** — six traders, three rounds, $b = 40$ — and thin markets like this are exactly where LMSR path-dependence is most likely to bite (caveat: thin markets). And the static baselines had access to *exactly the same information* as the market; that equality is the point of the design, not a handicap on the market.

4.2 H2a — mixed vs. homogeneous teams (replication)

H2(a): mixed-model teams outperform homogeneous teams.

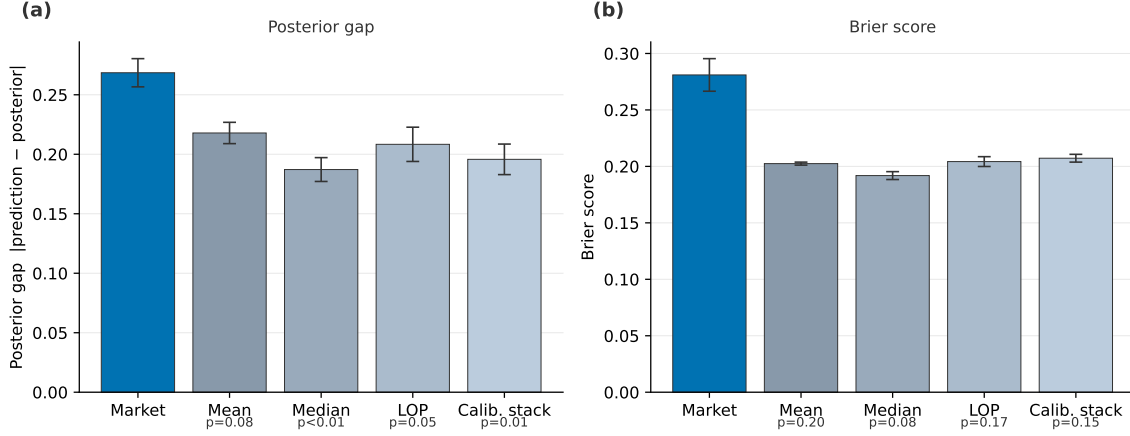


Figure 1: Posterior gap and Brier, market vs. static pools, computed over independently elicited beliefs from the same agents, with error bars.

This is a **replication** of the diversity direction reported for LLM prediction markets (arXiv:2606.26583), and it holds. The mixed team C has the smallest market posterior gap of any team (0.225 ± 0.021). Against the homogeneous llama arm B1 (0.380 ± 0.054) the improvement is large and highly significant: $\Delta = -0.155$ (CI $[-0.211, -0.100]$, $p < 0.001$). Against the homogeneous deepseek arm (condition A / B2) the gap is $\Delta = -0.044$ (CI $[-0.088, 0.000]$, $p = 0.106$), and against the homogeneous nemotron arm B3 (0.276 ± 0.028) it is $\Delta = -0.051$ (CI $[-0.107, 0.004]$, $p = 0.118$). The mixed team beats all three homogeneous arms in point estimate; the advantage is decisive against the weakest model and directional-but-not-significant against the two stronger ones. Because the three models were quality-matched on standalone gap ($0.291 / 0.268 / 0.257$), this advantage survives quality matching — it is composition, not a smuggled-in stronger model.

4.3 H2b — the phase map (descriptive)

H2(b): the market-vs-static gap shrinks as measured error correlation ρ rises.

For each of the 12 (condition, seed) points among A, B1, B3, and C we measured the team’s error correlation ρ from the Pass-1 beliefs and the market’s deficit relative to the mean pool (market gap – mean-pool gap). The Spearman rank correlation between ρ and the market deficit is $r_s = 0.280$ ($n = 12$). The sign is consistent with the hypothesis — higher correlation goes with a larger market deficit — and the low- ρ mixed team C is the **only** condition with per-seed deficits below zero (-0.010 and -0.029 on seeds 1 and 2), i.e. the only setting where the market actually helps. The relationship is far from clean, however: the team with the genuinely lowest error correlation is the homogeneous deepseek pool A (team-mean $\rho = 0.398$, and the single lowest- ρ seed at 0.378), and it still shows a market deficit on every seed ($+0.055 / +0.051 / +0.045$). That is exactly why the rank correlation is weak ($r_s = 0.280$) rather than strong — low correlation is necessary but not sufficient for the market to help in this environment.

This is **descriptive, not a fitted law** (caveat: narrow ρ range). The achieved ρ values span only 0.378 to 0.514 , a narrow band, and with $n = 12$ we report the rank correlation as a description of the map rather than as a hypothesis test. Fig 2 shows the phase map; the takeaway is a direction and a single below-zero region (the low-correlation mixed team), not a calibrated curve.

4.4 H3 — adversarial manipulation

H3: for an adversary with $k\times$ bankroll pushing the wrong side, report Brier degradation, flip rate, capital-per-move, adversary P&L, and recovery. Hypothesis: flip rate increases with k ; adversary P&L is negative at settlement.

We add one adversary to pool C, instructed to push the price toward the wrong side while staying plausible, at bankroll multiples $k \in \{1, 3, 10\}$. The manipulation works in the expected direction and grows with capital. The final-prediction **flip rate rises with k** : $0.317 \rightarrow 0.358 \rightarrow 0.392$. Terminal **Brier degradation** rises steeply: $0.105 \rightarrow 0.219 \rightarrow 0.312$. The share of questions pushed to the wrong side climbs from 0.508 to 0.617 to 0.658.

But the market **taxes** the attack. Adversary P&L at settlement is negative at every multiple, and the absolute loss grows with the capital deployed: $-\$46.60$ at $k=1$, $-\$143.06$ at $k=3$, and $-\$440.41$ at $k=10$. The mean capital required to move the price by 0.1 is $44.2 / 30.1 / 306.5$ across the three multiples. So while a well-funded adversary can flip a meaningful fraction of final predictions and degrade calibration, it pays progressively more to do so and never profits — consistent with the hypothesis that the market taxes manipulation.

Recovery is low: the post-attack recovery fraction is $0.158 / 0.200 / 0.092$ across k . An important **caveat**: recovery *timing* is structurally unmeasurable here. The adversary trades in every round, including the last, so there is no clean post-attack window in which the price could relax back; **median_recovery_rounds** is 0 by construction. We therefore report the recovery fraction only, not a recovery-time. The manipulation panel of Fig 2 shows flip rate and recovery fraction versus k .

4.5 H-wealth — persistent wealth (secondary)

H-wealth: persistent wealth across questions changes market accuracy vs. per-question reset; direction unknown; test order-dependence.

Letting bankrolls persist across questions **slightly worsens** the market’s posterior gap relative to the per-question reset of condition A. With a fixed question order, $W_FIXED - A$ is $\Delta = 0.050$ (CI $[-0.001, 0.098]$, $p = 0.035$); with a shuffled order, $W_SHUFFLED - A$ is $\Delta = 0.058$ (CI $[0.007, 0.111]$, $p = 0.032$). Both are small and marginally significant. There is **no order-dependence**: W_FIXED vs. $W_SHUFFLED$ gives $\Delta = -0.008$ (CI $[-0.050, 0.034]$, $p = 0.737$), so the effect does not depend on the sequence in which questions are seen. The Brier comparisons for the wealth conditions are not significant ($p \geq 0.39$). The direction — persistence hurts a little — was not pre-specified; we report it as found.

4.6 Secondary reporting

Parse reliability was high across the grid: 5 parse failures total (4 from deepseek, 1 from nemotron) out of tens of thousands of LLM turns, with per-model rates at or below 0.19% in every condition (Table 2). The market maker’s subsidy stayed within its theoretical worst-case bound: the maximum per-question subsidy observed was 27.726, just under the $b \cdot \ln 2 \approx 27.73$ ceiling at $b = 40$ (Table 2). Accuracy-per-dollar figures are in Table 1; because absolute spends are tiny and some arms ran on a free tier, these ratios are noisy and we report them for completeness rather than as a headline. Total grid spend was \$0.7255.

5. Why does the market hurt here?

This section is an **interpretation**, clearly labeled as such — a set of mechanisms consistent with the data, not a claim of proof.

Three ingredients plausibly combine to make LMSR trading lose information relative to independent pooling in this regime. First, **price commitment on the realized side**. A pilot observation (not itself a confirmed finding) was telling: at $b = 40$ the market’s final-price Brier beat the mean pool while its posterior gap was slightly worse — the market commits harder toward the side that turns out correct, which flatters the Brier against the realized outcome but overshoots the calibrated posterior. Overcommitment is penalized by the posterior-gap metric precisely because that metric knows the exact right answer, not just the realized coin flip. A smoke observation on held-out easy questions during the main runs showed the same pattern: market Brier can beat the pools while posterior gap does not — the market overcommits.

Second, **herding on shared errors**. When agents’ errors are correlated, trading lets them reinforce a common mistake: a price that already reflects the shared bias invites more trades in the same direction. The phase map (§4.3) is weakly consistent with this — the only condition whose market deficit ever turns negative is the low- ρ mixed team C, though the tendency is loose (the lowest- ρ homogeneous team, deepseek pool A, still runs a deficit on every seed). A static pool cannot amplify a shared error the way a sequential price can, because it never shows one agent another agent’s move.

Third, **LMSR path-dependence with thin participation**. With only six traders over three rounds, the final price depends on who traded when; there is no law of large numbers to wash out order effects, and the reshuffling that we do each round cannot fully average away path-dependence at this scale. Thin markets are where this mechanism is most acute.

None of these is established as *the* cause; they are the interpretations most consistent with the measured pattern, offered to structure follow-up work.

6. Extension: the capability axis

This section is an **exploratory extension**, run on 2026-07-24 after the initial release and **not part of the pre-registration**: the hypotheses here were formed after we saw the v1 reversal, and we report them as post-hoc.

Motivation. The reversal of §4.1 was measured on one trader model (**deepseek/deepseek-v4-flash**, “flash”). A natural worry is that it is an artifact of a *cheap* model — perhaps a weak trader mishandles prices and destroys information that a more capable trader would preserve. We test that by re-running condition A’s two-pass protocol on the same 40 questions $\times$ 3 seeds with two more capable homogeneous tiers.

Design. Three homogeneous tiers, identical questions, seeds, personas, and $b = 40$: **flash** (condition A), **pro** (A_PRO, **deepseek-v4-pro**), and **luna** (A_LUNA, **gpt-5.6-luna**). Pro is the key control — it is the *same model family* as flash, so it holds model knowledge roughly fixed while raising capability, and any change it produces on the market side is trading skill, not new knowledge. Luna is a stronger frontier model from a different vendor. Each tier’s pool gap, market gap, and deficit (market gap – mean-pool gap, positive $\Rightarrow$ the market destroys information) are computed exactly as in §4. The extension cost \$3.06 (\$0.45 for A_PRO, \$2.61 for A_LUNA).

Tier	Pool gap	Market gap	Deficit (market – pool)	Market Brier	Team ρ
flash	0.218	0.269	0.051 [−0.001, 0.100] (p = 0.080)	0.281	0.398 ± 0.018
pro	0.227	0.224	−0.003 [−0.048, 0.041] (p = 0.377)	0.215	0.378 ± 0.029
luna	0.191	0.192	0.002 [−0.046, 0.050] (p = 0.451)	0.201	0.283 ± 0.005

(n = 120 per tier; deficit CIs are 95% bootstrap, p from Wilcoxon signed-rank.)

Claim 1 — the deficit reaches parity, not superiority. Flash’s market runs a deficit of 0.051 [−0.001, 0.100] (p = 0.080) against its own pool. At both capable tiers the deficit vanishes — pro −0.003 [−0.048, 0.041] (p = 0.377) and luna 0.002 [−0.046, 0.050] (p = 0.451), CIs straddling zero — so a more capable trader **stops destroying information**. But it does not go past parity: no tier’s market beats its own pool, and the pro and luna market gaps (0.224, 0.192) sit essentially on top of their pool gaps (0.227, 0.191). The capable-market story is parity, not superiority.

Claim 2 — the fix is trading skill, not knowledge. Pro’s independent pool is no better than flash’s: the pro – flash pool-gap difference is 0.010 [−0.011, 0.030] (Wilcoxon p = 0.289), indistinguishable from zero and if anything nominally worse. Yet pro’s *market* gap is lower than flash’s by $\Delta = -0.044$ [−0.098, 0.008] (p = 0.127). Because pro shares flash’s model family its knowledge is held roughly fixed, and the improvement surfaces only once the market runs — so the deficit fix points to how the capable model *trades*, not to what it *knows*.

Claim 3 — capability matters independently of ρ . One could instead explain the fix through error correlation — lower ρ , less herding, smaller deficit. But pro’s team ρ (0.378 ± 0.029) is statistically indistinguishable from flash’s (0.398 ± 0.018) while their deficit outcomes are opposite, so capability moves the market-vs-pool gap through a channel other than measured ρ . Luna does confound capability with lower correlation ($\rho = 0.283 \pm 0.005$, and its pool is also the strongest — luna – flash pool gap −0.027 [−0.050, −0.004], p = 0.000), but the same-family pro control isolates capability from ρ .

Mechanism (supporting, descriptive). The herding regression — §3’s sanctioned dynamics-only analysis, regressing each agent’s stated in-market belief on its own Pass-1 belief (b1) and the pre-trade price (b2) — offers a consistent, if modest, mechanism. The pooled price-weight coefficient b2 is 0.045 [0.028, 0.060] for flash, 0.010 [−0.000, 0.021] for pro, and 0.023 [0.011, 0.036] for luna: flash’s traders weight the running price more than either capable tier’s (clearly so versus pro; the flash–luna difference is smaller and the CIs overlap), with mean belief drift $|y - x1|$ of 0.037 for flash vs. 0.012 for pro, while own-information weight b1 stays near 1 in every tier. This is **descriptive and the effect is small** — all three coefficients are modest and the pro-vs-luna ordering is not distinguishable (overlapping CIs) — so it is supporting evidence for the herding interpretation of §5, not proof.

Fig 2-v2 ([analysis_out/v2/fig2_v2.png](#)) shows the phase map of §4.3 with pro and luna added; Fig 3 ([analysis_out/v2/fig3.png](#)) shows the per-tier deficit and the price-weight b2 by tier and round. The full tier and herding tables live in [analysis_out/v2/tiers.md](#) and [herding.md](#), regenerated with:

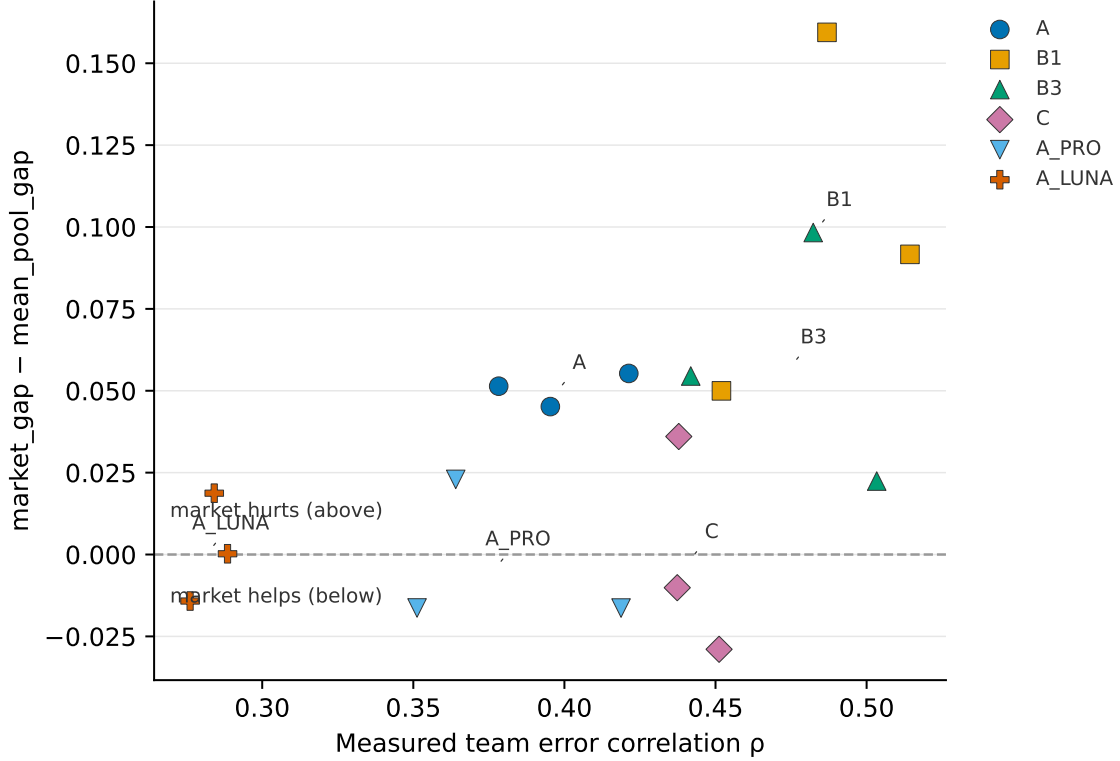


Figure 2: Phase map: market deficit vs. measured team error correlation ρ , with the pro and luna capability tiers added; manipulation panel of flip rate and recovery vs. adversary bankroll multiple k .

```
python -m pnyx.cli analyze-v2 --flash data/main/A --pro data/v2/A_PRO \
    --luna data/v2/A_LUNA --out analysis_out/v2
```

7. Limitations

We restate the binding caveats and add the design-scope ones.

1. **Recovery timing is structurally unmeasurable.** The adversary trades every round including the last, so there is no post-attack window; we report only the recovery fraction, never a recovery time (§4.4).
2. **The phase map is descriptive, not a fitted law.** The achieved team- ρ range is narrow (0.378–0.514) and $n = 12$, so the Spearman $r_s = 0.280$ is a description of direction, not a hypothesis test (§4.3).
3. **Thin markets.** All results are for six traders over three rounds — the regime where LMSR path-dependence and overcommitment are most likely to matter (§4.1, §5).
4. **Prompt and liquidity sensitivity.** b was chosen from a three-point pilot sweep and the

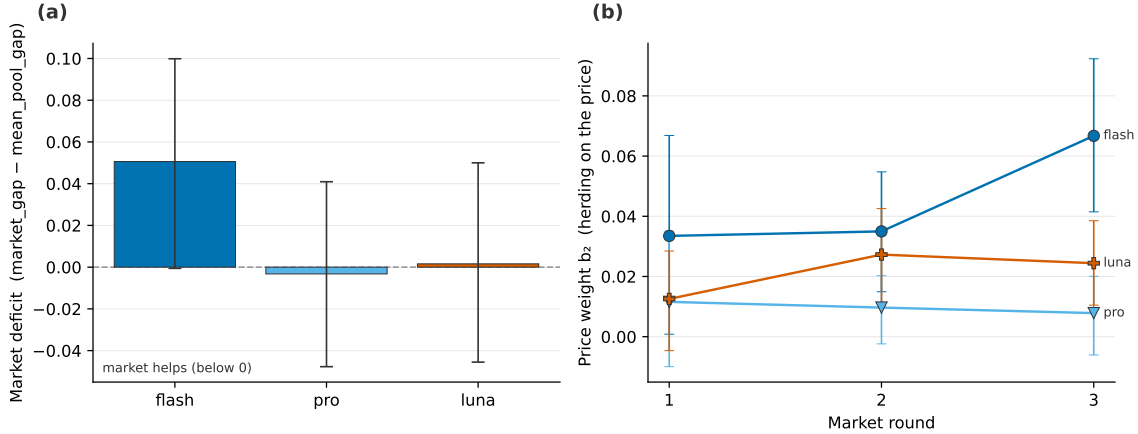


Figure 3: Per-tier market deficit (market gap minus pool gap) and the herding-regression price-weight b_2 , by capability tier and trading round.

prompts were frozen after a pilot, not exhaustively searched; different prompting could shift the market’s behavior (§3).

5. **Synthetic rendering.** Signals are rendered into template-constrained prose and adversarially verified, but this may not transfer to real documents with their noise, redundancy, and stylistic tells (§2).
6. **Single environment family.** Everything here is a binary latent state with six conditionally independent signals under a uniform prior; other signal structures, priors, or outcome cardinalities are untested (§2).
7. **The capability axis is exploratory and thin.** The §6 extension was not pre-registered — its hypotheses were formed after the v1 reversal — and it adds only two tiers. Luna is the single non-deepseek frontier vendor tested, so cross-vendor capability rests on one model; and luna confounds capability with lower error correlation, a confound the same-family pro control mitigates but does not remove. All three tiers ran on the same 40-question environment (§2), so the parity finding inherits every environment caveat above.

Two further points. The static baselines had access to *exactly the same information* as the market — that equality is the design, and it means the comparison is fair rather than that the market was disadvantaged. And every quantitative claim here is specific to this regime (6 agents, 3 rounds, $b = 40$, the configured signal structure); we make no claim that the reversal generalizes beyond it. Peer prediction — which could in principle distinguish genuine private information from correlated agreement — is deliberately out of scope for this version (arXiv:2601.20299) and left to a sequel.

8. Reproducibility

The pre-registration in SPEC.md fixes the hypotheses and the two-pass elicitation protocol before any main run. The P4 event logs are released under `data/main/`, and every figure, table, and statistic in this writeup regenerates from them with a single command and no API calls:

```
python -m pnyx.cli analyze-main --data-root data/main \
    --configs pnyx/configs/main --out analysis_out
```

The run completes in a few seconds. All bootstrap statistics are deterministic under a fixed boot-

strap seed, so the CIs and p-values above are reproducible exactly. Rerunning the experiments from scratch requires an OpenRouter key and costs about \$0.75 in total; the analysis of the released logs costs nothing.

9. Conclusion

In a controlled environment where the right answer is known exactly, LMSR market trading among a small pool of LLM agents recovers *less* of the available information than simply pooling those same agents’ independent beliefs — a reversal of the naive expectation, and the clearest single point on our help/nothing/hurt map. The market helps in only one measured corner: a low-correlation, quality-matched mixed team. Model diversity helps as prior work found (a replication, not a new claim), adversarial capital moves prices and degrades accuracy but is always taxed at settlement, and wealth persistence hurts slightly. A post-hoc capability extension refines the reversal rather than overturning it: cheap traders make LMSR destroy information, more capable traders — including a same-family control that isolates trading skill from knowledge — pull the market back to parity with the pool, and no tested tier’s market beats independent pooling. The value of the design is not any one of these numbers but the separation that produced them: mechanism, model knowledge, and signal structure pulled apart against an exact oracle, so that “the market helped” can be checked rather than assumed. The interpretation of *why* the market hurts here — overcommitment, herding on shared errors, thin-market path-dependence — is offered as structure for follow-up, not as settled fact.

10. References

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